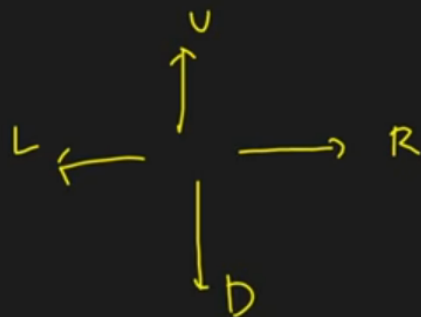


	0	1	2	3
0	1	0	0	0
1	1	1	0	1
2	1	1	0	0
3	0	1	1	1



$\checkmark$
DDRRRR < $\checkmark$
DRDDRR

L19. Rat in A Maze | Backtracking

2.20

0	1	0	0
1	1	0	1
2	1	1	0
3	0	1	1

✓			
✓			
✓	✓		
	✓	✓	

vis

$f(0,0, "")$ (D|L|R|U)

D L R U

$f(1,0, "D")$ D|L|R|U

DD R D R R

$f(2,0, "DD")$ ~~D~~|L|R|U

$f(2,1, "DDR")$ D|L|R|U

$f(3,1, "DDRD")$ ^xD|^xL|R|U

$f(3,2, "DDRD R")$ ^xD|^xL|[✓]R|U

$f(3,3) DDRDRR$



10:09 / 25:09



L19. Rat in A Maze | Backtracking

2.20

0	1	0	0
1	1	0	1
2	1	1	0
3	0	1	1

✓			
✓			
✓	✓		
	✓	✓	

vis

$f(0,0, "")$ (D|L|R|U)

D L R U

$f(1,0, "D")$ ~~D~~|L|R|U

DD R D R R

$f(2,0, "DD")$ ~~DD~~|L|R|~~U~~

$f(2,1, "DDR")$ ~~DDR~~|L|R|~~U~~

$f(3,1, "DDRD")$ ~~DDRD~~|L|R|~~U~~

$f(1,1, "DDRUD")$ ~~DDRUD~~|L|R|~~U~~

$f(3,2, "DDRRD")$ ~~DDRRD~~|L|R|~~U~~

$f(3,3, "DDRRDR")$



13:08 / 25:09



L19. Rat in A Maze | Backtracking

2.20

0	1	0	0
1	1	0	1
2	1	0	0
3	0	0	1

✓			
✓	✓		
✗	✓		
	✗	✗	

vis

$f(0,0, "")$ (D/L/R/U)

D L R U

$f(1,0, "D")$ D/L/R/U

$\begin{cases} \text{DDRDRR} \\ \text{DRDDRR} \end{cases}$

$f(1,1, "DR")$ D/L/R/U

$f(2,1, "DRD")$ D/L/R/U

$f(3,1, "DRDD")$ ✗/✗/✓/✓/✓/✓

$f(3,2, "DRDDR")$ ✗/✗/✓/✓/✓/✓

$f(3,3, "DRDDRR")$

$f(2,0, "DD")$ ✗/✗/✓/✓

$f(2,1, "DDR")$ ✗/✗/✓/✓/✓

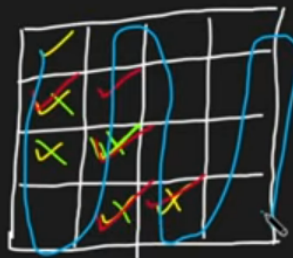
$f(1,1, "DDR")$ ✗/✗/✓/✓/✓

$f(3,1, "DDR")$ ✗/✗/✓/✓/✓

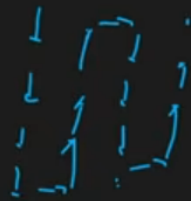
$f(3,2, "DDRDR")$ ✗/✗/✓/✓/✓/✓

$f(3,3, "DDRDRR")$

	0	1	2	3
0	1	0	0	0
1	1	1	0	1
2	1	1	0	0
3	0	1	0	1



vis



$f(0,0, "")$ (D|L|R|U)

$f(1,0, "D")$ D|L|R|U

$f(2,0, "DD")$ D|L|R|U

$f(2,1, "DDR")$ D|L|R|U

$f(3,1, "DDRD")$ D|L|R|U

$f(3,2, "DDRRD")$ D|L|R|U

$f(3,3, "DDRRDR")$

$f(1,1, "DDR")$ D|L|R|U

$f()$

Down

TC $\rightarrow 4^{(n \times m)}$
SC $\rightarrow O(m \times n)$

D L R U

$\begin{bmatrix} DDRDRR \\ DRDDRR \end{bmatrix}$

$f(1,1, "DR")$ D|L|R|U

$f(2,1, "DRD")$ D|L|R|U

$f(3,1, "DRDD")$ D|L|R|U

$f(3,2, "DRDDR")$ D|L|R|U

$f(3,3, "DRDDRR")$

	X	X	

vis

$f(2,1, DDR)$ $\overline{D/L/R/U}$

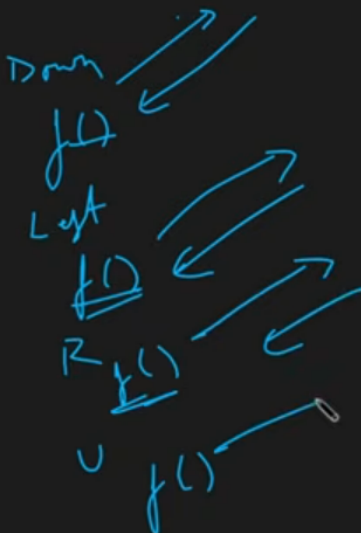
$f(3,1, DDRD)$ $\overline{D/L/R/U}$

$f(3,2, DDRDR)$ $\overline{D/L/R/U}$

$f(3,3, DDRDRR)$

$f(1,1, DDRU)$ $\overline{D/L/R/U}$

$f()$



$f(2,1, DDR)$ $\overline{D/L/R/U}$

$f(3,1, DDRD)$ $\overline{D/L/R/U}$

$f(3,2, DDRDR)$ $\overline{D/L/R/U}$

$f(3,3, DDRDRR)$

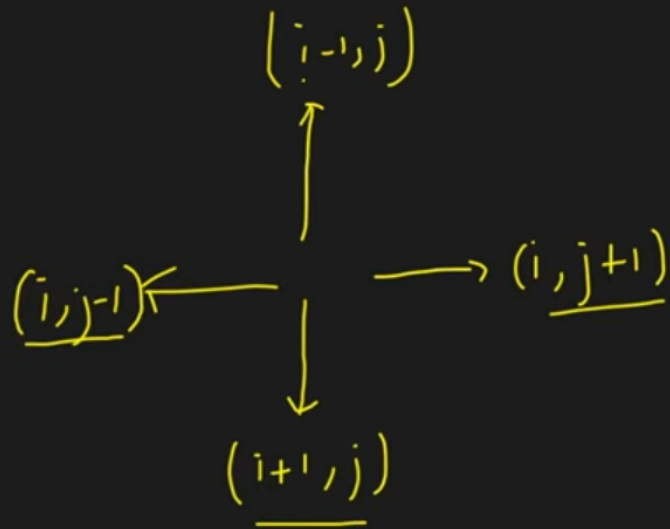
L19. Rat in A Maze | Backtracking

```
11 class Solution{
12     void solve(int i, int j, vector<vector<int>> &a, int n, vector<string> &ans, string move, vector<vector<int>> &vis) {
13         if(i==n-1 && j==n-1) {
14             ans.push_back(move);
15             return;
16         }
17         // downward
18         if(i+1<n && !vis[i+1][j] && a[i+1][j] == 1) {
19             vis[i][j] = 1;
20             solve(i+1, j, a, n, ans, move + 'D', vis);
21             vis[i][j] = 0;
22         }
23
24         // left
25         if(j-1>=0 && !vis[i][j-1] && a[i][j-1] == 1) {
26             vis[i][j] = 1;
27             solve(i, j-1, a, n, ans, move + 'L', vis);
28             vis[i][j] = 0;
29         }
30
31         // right
32         if(j+1<n && !vis[i][j+1] && a[i][j+1] == 1) {
33             vis[i][j] = 1;
34             solve(i, j+1, a, n, ans, move + 'R', vis);
35             vis[i][j] = 0;
36         }
37
38         // upward
39         if(i-1>=0 && !vis[i-1][j] && a[i-1][j] == 1) {
40             vis[i][j] = 1;
41             solve(i-1, j, a, n, ans, move + 'U', vis);
42             vis[i][j] = 0;
43         }
44     }
45 public:
46     vector<string> findPath(vector<vector<int>> &m, int n) {
47         vector<string> ans;
```

```

15     return;
16 }
17 // downward
18 if(i+1<n && !vis[i+1][j] && a[i+1][j] == 1) {
19     vis[i][j] = 1;
20     solve(i+1, j, a, n, ans, move + 'D', vis);
21     vis[i][j] = 0;
22 }
23
24 // left
25 if(j-1>=0 && !vis[i][j-1] && a[i][j-1] == 1) {
26     vis[i][j] = 1;
27     solve(i, j-1, a, n, ans, move + 'L', vis);
28     vis[i][j] = 0;
29 }
30
31 // right
32 if(j+1<n && !vis[i][j+1] && a[i][j+1] == 1) {
33     vis[i][j] = 1;
34     solve(i, j+1, a, n, ans, move + 'R', vis);
35     vis[i][j] = 0;
36 }
37
38 // upward
39 if(i-1>=0 && !vis[i-1][j] && a[i-1][j] == 1) {
40     vis[i][j] = 1;
41     solve(i-1, j, a, n, ans, move + 'U', vis);
42     vis[i][j] = 0;
43 }
44 }
45 public:
46 vector<string> findPath(vector<vector<int>> &m, int n) {
47     vector<string> ans;
48     vector<vector<int>> vis(n, vector<int>(n, 0));
49     if(m[0][0] == 1) solve(0,0,m,n, ans, "", vis);
50     return ans;
51 }
52 };
53

```

D | L | R | U

	$\uparrow$	$\rightarrow$	$\downarrow$	$\leftarrow$
$di[] =$	+1	+0	+0	-1
$dj[] =$	+0	-1	+1	+0
	D	L	R	U

$$\Rightarrow i + di[\text{mel}]$$

$$\Rightarrow j + dj[\text{mel}]$$

```

10
11 class Solution{
12     void solve(int i, int j, vector<vector<int>> &a, int n, vector<string> &ans, string move,
13     vector<vector<int>> &vis, int di[], int dj[]) {
14         if(i==n-1 && j==n-1) {
15             ans.push_back(move);
16             return;
17         }
18         string dir = "DLRU";
19         for(int ind = 0; ind<4;ind++) {
20             int nexti = i + di[ind];
21             int nextj = j + dj[ind];
22             if(nexti >= 0 && nextj >= 0 && nexti < n && nextj < n && !vis[nexti][nextj] && a[nexti][nextj] == 1) {
23                 vis[i][j] = 1;
24                 solve(nexti, nextj, a, n, ans, move + dir[ind], vis, di, dj);
25                 vis[i][j] = 0;
26             }
27         }
28     }
29 public:
30     vector<string> findPath(vector<vector<int>> &m, int n) {
31         vector<string> ans;
32         vector<vector<int>> vis(n, vector<int> (n, 0));
33         int di[] = {+1, 0, 0, -1};
34         int dj[] = {0, -1, 1, 0};
35         if(m[0][0] == 1) solve(0,0,m,n, ans, "", vis, di, dj);
36         return ans;
37     }
38 };
39
40
41
42 // } Driver Code Ends

```